RASPBERRY PI LAB

mmm...Pi

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INTRODUCTION

OBJECTIVES:

- Image an SD card with Raspbian OS using the Raspberry Pi Imager.
- Pre-configure network information and SSH access.

Host Name	**Root User**
davidPi	david

- Boot your Raspberry pi and connect to it through PuTTY.
- Update Raspbian OS.
- Configure three user accounts and give each of them ownership of a specific folder

  User accounts: daveon, dAvis, deejay

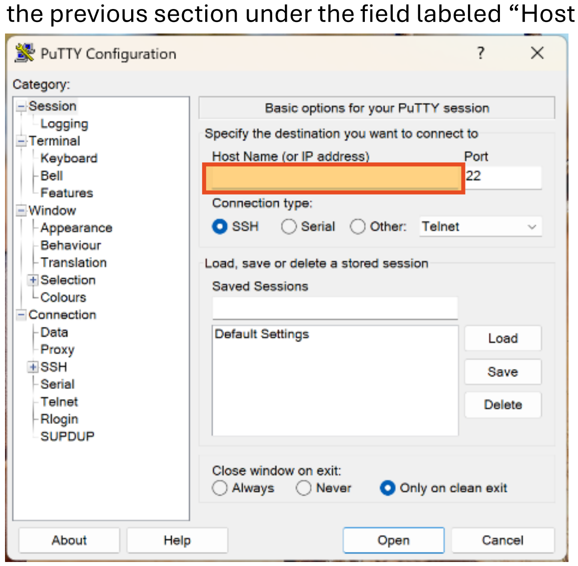
BOOTING THE PI AND FINDING THE IP

Once the Raspberry Pi has booted with the image written to the MicroSD card, open the command prompt on the laptop. Use the ping command with the host name that was preconfigured using the Raspberry Pi Imager. Make note of the resulting IP address.

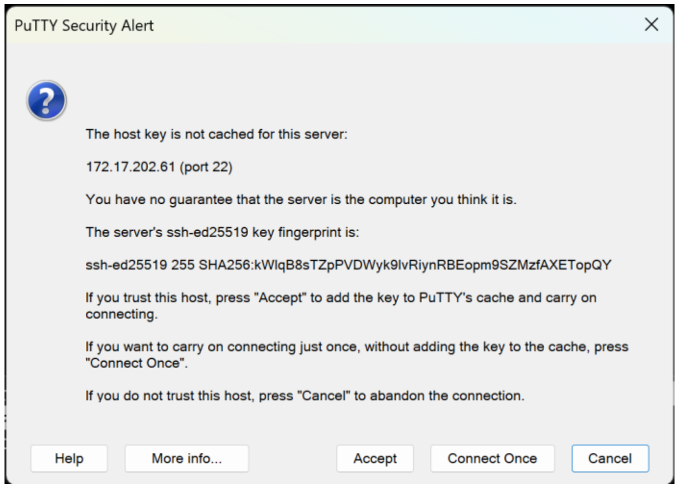
```
C:\Users\TFLaptop029>ping davidPi.local -4

Pinging davidPi.local [172.17.202.61] with 32 bytes of data:
Reply from 172.17.202.61: bytes=32 time=208ms TTL=64
Reply from 172.17.202.61: bytes=32 time=14ms TTL=64
Reply from 172.17.202.61: bytes=32 time=9ms TTL=64
Reply from 172.17.202.61: bytes=32 time=9ms TTL=64

Ping statistics for 172.17.202.61:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 9ms, Maximum = 208ms, Average = 60ms

C:\Users\TFLaptop029>
```

CONNECTING TO YOUR PI THROUGH SSH

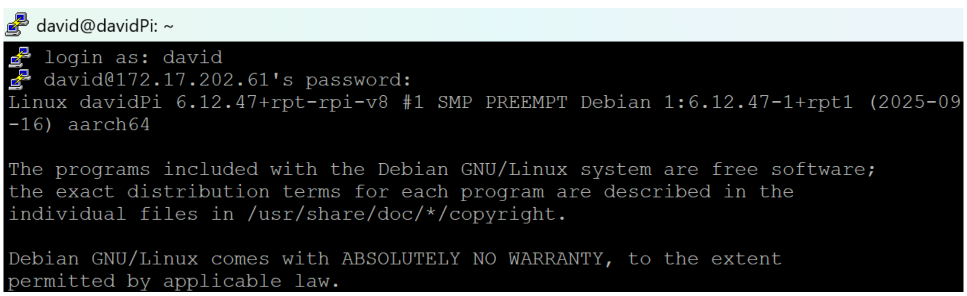
1. Open the putty application
2. Enter the IP address from the previous section under the field labeled “Host Name (or IP address)”.



3. Click Open. A window will appear.



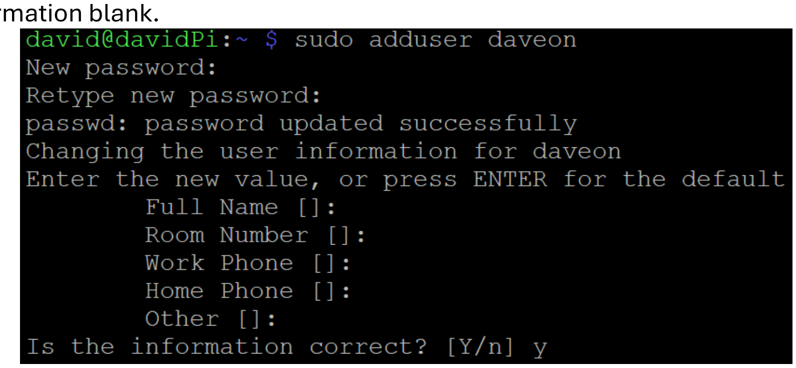
4. Click “Accept”. You will be prompted to “login as”
5. Using the login information that was preconfigured using the Raspberry Pi Imager, enter the username then password.



ADDING ACCOUNTS AND FOLDERS

While connected to the Raspberry Pi through SSH:

1. Enter “sudo adduser [new username]” command to create a new user
2. Enter the new user’s password
3. Press enter 5 times. Here we would enter the user’s information. In this instance we will be leaving the user’s information blank.



4. Press “y” to confirm

5. Repeat steps 1 through 4 for each user.
    - Alternatively we can forego entering the user's information and the confirmation, by replacing the command in step 1 with "sudo adduser --gecos "" [new username]". Skipping steps 3 and 4.

```
david@davidPi:~ $ sudo adduser --gecos "" dAvis
New password:
Retype new password:
passwd: password updated successfully
usermod: no changes
david@davidPi:~ $
```

6. Enter the "cd .." command until only "[your username]@[hostname]:/ $" is visible before your cursor to ensure you are in the root directory
7. Enter "sudo mkdir storage" to create a directory called storage
8. Enter "sudo mkdir [username]" for each user created previously.

```
david@davidPi:/storage $ sudo mkdir daveon
david@davidPi:/storage $ sudo mkdir deejay
david@davidPi:/storage $ sudo mkdir dAvis
david@davidPi:/storage $ ls -la
total 20
drwxr-xr-x  5 root root 4096 Mar 18 15:17 .
drwxr-xr-x 19 root root 4096 Mar 18 15:08 ..
drwxr-xr-x  2 root root 4096 Mar 18 15:16 daveon
drwxr-xr-x  2 root root 4096 Mar 18 15:17 dAvis
drwxr-xr-x  2 root root 4096 Mar 18 15:17 deejay
david@davidPi:/storage $
```

9. Enter "sudo chown [username] [username]" to change the owner of the directory to the corresponding user. The first [username] is the target owner and the second [username] is the target directory. Repeat this step for user created previously

```
david@davidPi:/storage $ sudo chown daveon daveon
david@davidPi:/storage $ sudo chown dAvis dAvis
david@davidPi:/storage $ sudo chown deejay deejay
david@davidPi:/storage $ ls -la
total 20
drwxr-xr-x  5 root   root 4096 Mar 18 15:17 .
drwxr-xr-x 19 root   root 4096 Mar 18 15:08 ..
drwxr-xr-x  2 daveon root 4096 Mar 18 15:16 daveon
drwxr-xr-x  2 dAvis  root 4096 Mar 18 15:17 dAvis
drwxr-xr-x  2 deejay root 4096 Mar 18 15:17 deejay
david@davidPi:/storage $
```